

Ratio and Proportion: Bar Models

Using Bar Models to Solve Problems

Ratio problems can be solved using diagrams.

Let's say we are solving the following problem:

$\frac{6}{8}$ of the vehicles on Joe's street are cars. If 24 vehicles are cars, how many vehicles are there in total?

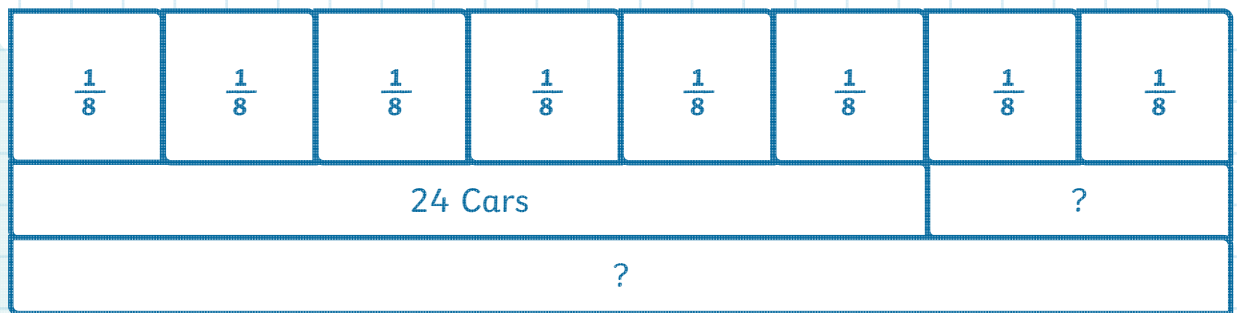
We can draw a bar model to help.

We know that $\frac{6}{8}$ of the vehicles are cars, and that there are 24 cars.

We don't know how many vehicles are not cars yet.

But we do know that it is the remaining $\frac{2}{8}$.

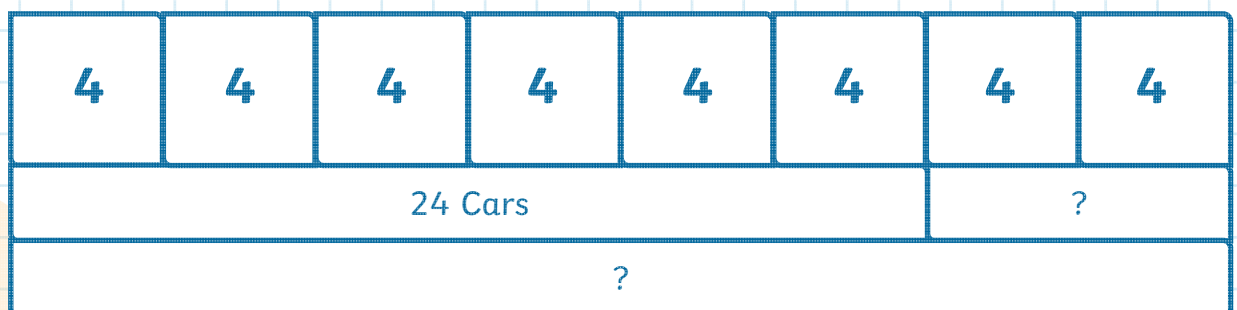
vehicles



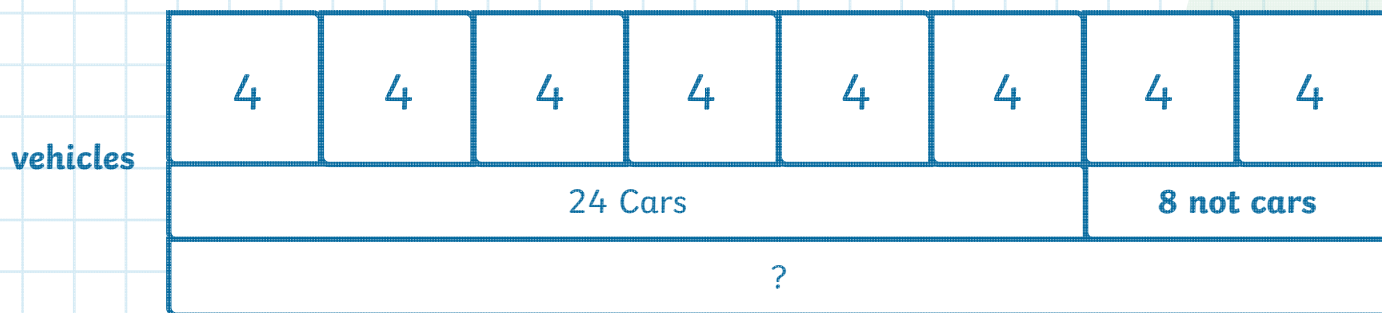
We can solve this by finding out what $\frac{1}{8}$ is.

Since we know that $\frac{6}{8} = 24$, we can find $\frac{1}{8}$ by dividing 24 by 6. $24 \div 6 = 4$,
so $\frac{1}{8} = 4$.

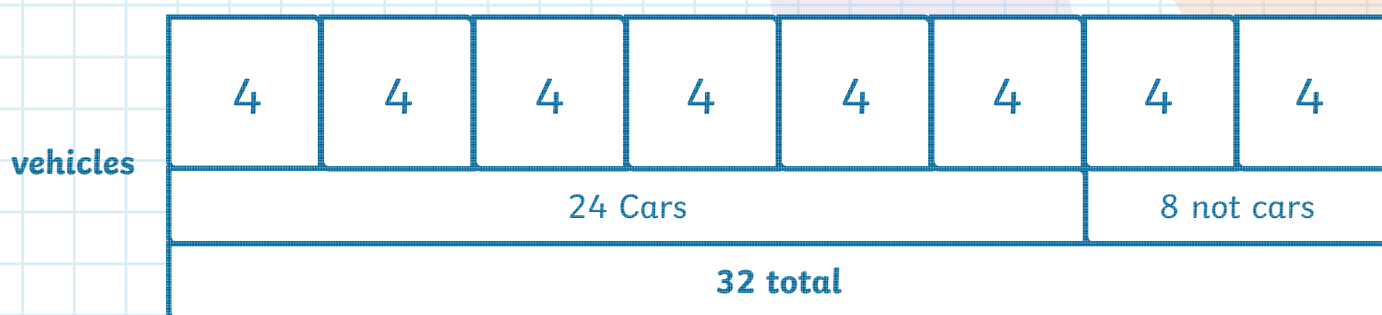
vehicles



Now, we can use the bar model to find out what proportion of the vehicles are not cars.



Now, we have enough information to find out how many vehicles there are in total on Joe's street.



Ratio and Proportion: Unequal Grouping

We can also use bar models to solve problems with unequal groups.

For example, let's say we are solving the following problem:

The cafe is having a sale. For every 3 hot drinks you buy, you get a free cake. How many hot drinks would you need to buy to get 7 free cakes?

We can draw a bar model to help. We know that for every 3 drinks, you get 1 cake.

Hot drink	1	2	3
Free cake	1		

We can keep on counting along in this way until we get to 7 free cakes.

Hot drink	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
Free cake	1			2			3			4			5			6			7		

This bar model also shows us that the ratio of hot drinks to cakes is 3:1.

If you feel confident, you could even show it like this:

Hot drink	3	6	9	12	15	18	21
Free cake	1	2	3	4	5	6	7

We would need to buy 21 hot drinks to get 7 free cakes.

Ratio and Proportion: Unequal Sharing

We can use bar models to solve problems with unequal sharing too.

Altogether, Amelia and Ezra are given £33 pocket money.

However, Amelia did more chores than Ezra, so she gets £3 more than him.

How much pocket money does each child get?

We can take £3 from the total of £33 first and set that aside for Amelia.

£33	
£30	£3
	Amelia

Now we can split the remaining £30 equally between Amelia and Ezra.

£33	
£30	£3
Ezra	Amelia
£15	£18

Ezra earned £15 and Amelia earned £18.

£18 is £3 more than £15, and $£18 + £15 = £33$, so we know that we are correct.

Twinkl Tip!

Another way we can work this out is by first dividing the total between Ezra and Amelia, giving them £16.50 each. We know that Amelia has £3 more than Ezra. There are two of them, so dividing £3 by 2 is £1.50. Take £1.50 from Ezra and add £1.50 to Amelia to find out how much they each earned.

£30	
Ezra $£16.50 - £1.50$ £15	Amelia $£16.50 + £1.50$ £18

Problems

Solve the following ratio and proportion problems using bar models.

1. In Zainab's class, $\frac{3}{5}$ children wear a tie. If 18 children wear a tie, how many children are in the class?

2. Marek is painting. He is 60% of the way through his tin of paint. If he has used 240 ml of paint already, how much paint can the tin hold?

3. 70% of the crowd at the football match are Blues supporters. If 112 people are Blues supporters, how many people are in the crowd?

4. For every 8 streets that he walks, Rex the dog gets 2 treats. How many treats would Rex get if he walked...?

16 streets

44 streets

What is the ratio of streets walked to treats given?

5. Mina is doing some baking. Her recipe calls for 300g of sugar for every 200g of flour. How much sugar will she need if she's using...?

800g of flour

1kg of flour

What is the ratio of sugar to flour?

6. To make cordial, I need 3 drops of flavouring per 200 ml of water. How many drops will I need for...?

600ml

1.5 litre

What is the ratio of flavouring to water?

7. Ling and Arthur have cycled a total of 45 miles. Ling cycled 5 miles farther than Arthur. How far did they each cycle?

Ling:

Arthur:

8. Timo and Lilly are hungry dogs. Every day, they eat 600g of dog food between them. Lilly eats 120g more than Timo each day. How much do they both eat?

Timo:

Lilly:

9. Connor and Nick have earned £72 on their new song so far. Connor did the majority of the songwriting, so he has earned £12 more than Nick. How much did they each earn?

Connor:

Nick:

10. In total, Roman, Sarah and Gorka scored 418 points in the video game tournament. Gorka scored 34 points more than Roman and Sarah, and Sarah scored 6 points more than Roman. How much did they each score?

Roman:

Sarah:

Gorka:

Answers

1. In Zainab's class, $\frac{3}{5}$ children wear a tie. If 18 children wear a tie, how many children are in the class?

$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$	$\frac{1}{5}$
6	6	6	6	6
18 tie			12 no tie	
30 total				

30

2. Marek is painting. He is 60% of the way through his tin of paint. If he has used 240 ml of paint already, how much paint can the tin hold?

10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
40ml	40ml	40ml	40ml	40ml	40ml	40ml	40ml	40ml	40ml
240ml used						160ml left			
400ml total									

400ml

3. 70% of the crowd at the football match are Blues supporters. If 112 people are Blues supporters, how many people are in the crowd?

10%	10%	10%	10%	10%	10%	10%	10%	10%	10%
16	16	16	16	16	16	16	16	16	16
112 Blues							48 non-Blues		
160 total									

160 total

4. For every 8 streets that he walks, Rex the dog gets 2 treats. How many treats would Rex get if he walked...?

16 streets

4

44 streets

11

What is the ratio of streets walked to treats given?

8:2 or 4:1

Streets	8		16		24		32		40		48	
Treats	1	2	3	4	5	6	7	8	9	10	11	12

5. Mina is doing some baking. Her recipe calls for 300g of sugar for every 200g of flour. How much sugar will she need if she's using...?

800g of flour

1200g

1kg of flour

1500g

What is the ratio of sugar to flour?

300g:200g or 3:2

Flour (g)	200			400			600			800			1000		
Sugar (g)	100	200	300	400	500	600	700	800	900	1000	1100	1200	1300	1400	1500

6. To make cordial, I need 3 drops of flavouring per 200 ml of water. How many drops will I need for...?

600ml

9

1.5 litre

23.5 or $23\frac{1}{2}$

What is the ratio of flavouring to water?

3 drops:200ml

Water (ml)

Flavouring
(drops)

200			400			600			800			1000			1200			1400			1600		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24

7. Ling and Arthur have cycled a total of 45 miles. Ling cycled 5 miles farther than Arthur. How far did they each cycle?

Ling: **25 miles**

Arthur: **20 miles**

45 miles		
40		5
20	20	5
Arthur	Ling	

8. Timo and Lilly are hungry dogs. Every day, they eat 600g of dog food between them. Lilly eats 120g more than Timo each day. How much do they both eat?

Timo: **240g**

Lilly: **360g**

600g		
480g		120g
240g	240g	120g
Timo	Lilly	

9. Connor and Nick have earned £72 on their new song so far. Connor did the majority of the songwriting, so he has earned £12 more than Nick. How much did they each earn?

Connor: **£42**

Nick: **£30**

£72		
£60		£12
£30	£30	£12
Nick	Connor	

10. In total, Roman, Sarah and Gorka scored 418 points in the video game tournament. Gorka scored 34 points more than Roman and Sarah, and Sarah scored 6 points more than Roman. How much did they each score?

Roman: **126**

Sarah: **132**

Gorka: **160**

418				
378			34	6
126	126	126	34	6
Roman	Sarah	Gorka	Gorka	6

We hope you find the information on our website and resources useful. As far as possible, the contents of this resource are reflective of current professional research. However, please be aware that every child is different and information can quickly become out of date. The information given here is intended for general guidance purposes only and may not apply to your specific situation.